K-CAP PROJECT

Krittika Computational Astronomy Project

Solar Flare Detection

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Automated Flare Detection System

Abstract

Solar flares are powerful and energetic explosions occurring on the Sun's surface, resulting in the release of a broad spectrum of electromagnetic radiation across space. These explosive events, which can release enormous amounts of energy, are highly dynamic and can vary in intensity, lasting anywhere from a few minutes to several hours. The frequency and intensity of solar flares are closely linked to the Sun's 11-year solar cycle, which consists of periods of solar minimum and solar maximum. When the solar cycle is at a minimum, active regions are small and rare and few solar flares are detected. These increase in number as the Sun approaches the maximum part of its cycle.

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1 Introduction

There aren't a great number of ways of identifying the stochastic emissions of the Sun known as solar flares, as of now. Furthermore, the process of detecting multiple solar flares in rapid succession, or multi-flare groups, remains a difficult task, as current detection methods often struggle to handle such complex scenarios.

This paper focuses on addressing these challenges by developing an optimal automated algorithm capable of detecting solar flares from data received by space-based observatories. Specifically, we aim to create a method that can identify solar flare events accurately and efficiently, even in the presence of noise or disturbances in the data. We analyze the data recorded by Aditya L1 in the month of December, providing real-world data to assess the algorithm's performance and potential for future flare detection systems.

Objective

Our objective is to develop an automated algorithm for solar flare detection that can accurately identify solar flares, including their start and end times, as well as their durations.

The algorithm should be capable of filtering out small disturbances and noise present in the data to focus on the true flare events. Additionally, we aim to fit the processed data to a curve that represents the flare characteristics, ensuring that the curve provides a high-quality fit to the data. This fitted curve should effectively highlight the peaks in the light curve, providing clear and precise visual representations of solar flare activity.



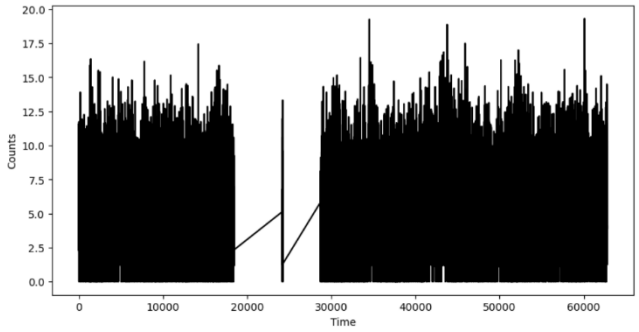
2. Algorithm Description

Smoothing the data

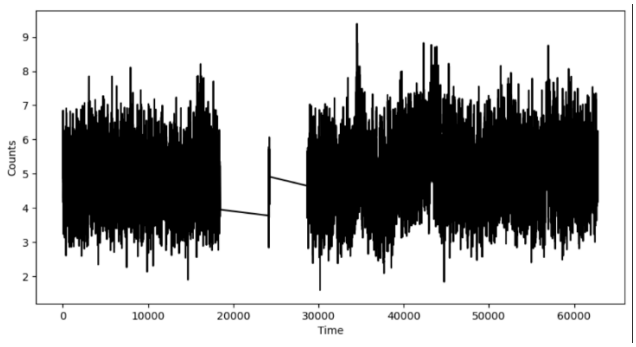
Binning

We take the data collected by Aditya L1 as our input for determining the presence of solar flares. The counts of solar flares were recorded with the time during which they were said to occur. The raw data is initially full of noise and needs to be binned to somewhat accurately sort out the unwanted elements of the flare data. We set the bin size initially to 120 seconds. Keeping the bin size too large will result in the loss of resolution and make it difficult to identify important features like peaks, flares, or sharp transitions in our data.

We extend the arrays on both sides by a size of bin_size//2 using edge padding then apply convolution. This way we get smoothed data removing the unnecessary noises and disturbances. We also set the time to scale by subtracting all the time values from the first recorded time value.



(a)



(b)

Graphs indicate the data points collected by Aditya L1 for the date 12th December 2019. (a) The original data points plotted with time (b) The data points binned with a kernel size of 10s plotted with time.

Filters

One can apply various filters to help smooth out the data. Commonly used filters are the median filter, the gaussian filter and the mean filter. The implementation of the mean and the median filter can again be carried out in a number of ways either using normal Python functions or the library numpy.

Interpolation

The raw data we have at hand contains missing values. These missing values in solar flare data arise from a combination of technical, environmental, and observational limitations. Researchers and engineers work to minimize these gaps, but they are a common part of working with real-world scientific data. We go on to identify these missing points and interpolate their values. We create an array ‘full_time’ containing continuous time values in seconds. We then append the value ‘Nan’ at those time points where the data wasn’t recorded and interpolate these values to get smooth data points.

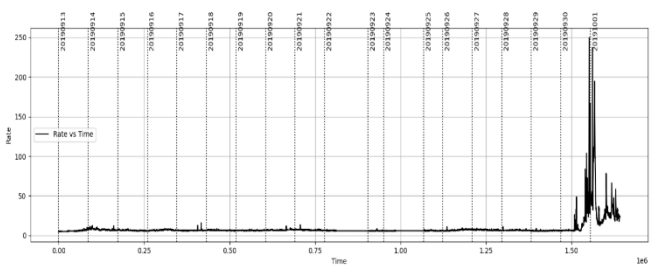


Figure 2.1.1. Data recorded for the month of December in the year 2019 binned with a kernel size of 120s.

Background estimation

We begin by getting a vague estimate of our background for the data. The data for solar flares usually has a lot of background solar activities. The data provided by Aditya L1 does not provide us with the background estimate and this is important to study the flares accurately. The solar activities by the Sun constantly emit X-ray photons and contribute to the intensity and isolating the flare from this background allows for a better understanding of the flare's intensity,

duration, and energy. We define the term "flaring duration" to accurately get the background of the data. Flaring durations are time stamps where a flaring activity takes place. We determine the start and end of the flares and get their moment of occurrence to get the duration of a flare. "Non-Flaring durations" are defined as the regions other than the flaring durations in the data.

Here, we run into an issue. In order to accurately determine the background of some data it is essential to obtain the flaring duration then subtract this duration from the original data to get the non-flaring duration. But in order to determine the non-flaring duration accurately we need to subtract the background estimate from it. This leads us to a circular problem.

We debug this by vaguely selecting the flaring durations, getting the non-flaring durations, determining the background then getting back the flaring durations with the subtracted background. The core idea is to model the non-flaring background, detect deviations from this background, and then refine the detection over multiple iterations to reduce false positives that is the noise and false negatives that is the missed flares.

Flare identification

Our goal with this project is to create an automated flare detection system. We start with identifying the flares by detecting the start and end times of each flare. We then mark all the prominent peaks from the data using the library from scipy.signal import find_peaks. We set the prominence to around 8-10 to identify all the large flares. In short, using prominence allows for cleaner and more meaningful peak detection by filtering out noise and focusing on the most significant and relevant features in the data.

After marking all the peaks, we gradually slide down from each peak by iterating and determine the slope at every point. Each flare would need the slope at its starting and ending points to be close to zero as the data points would need to rise again after the end of one flare. We set up a threshold value close to zero and iterate through the peaks marked.

We obtain the index of each peak then in succession decrease the index till we obtain an index where the value of the data divided upon the value of time elapsed till that point turn up to be close to zero. We mark this point as the starting point of the flare and repeat the same idea to obtain

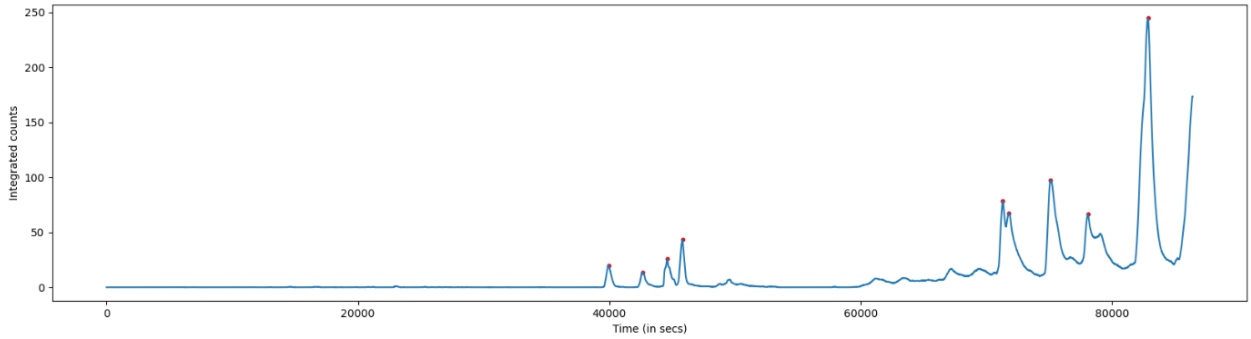


Figure 2.3.1

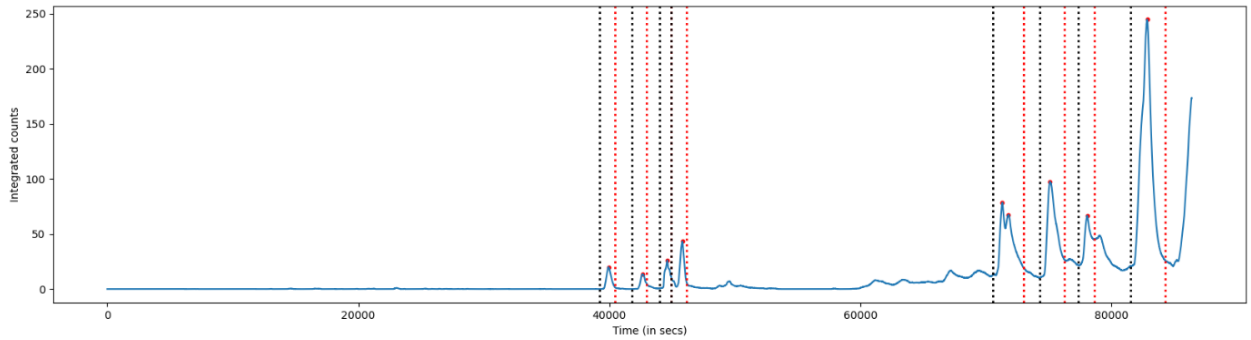


Figure 2.3.2

Background-subtracted solar flares data collected for the date 30th December 2019. Peaks marked with red dots along with the start and end time of all the flares. Figure 2.3.1 shows the peaks marked. while Figure 2.3.2 shows the start and end times of each flare. The black dotted vertical lines indicate the start of a flare while the red dotted vertical lines indicate the end of a flare.

the ending point by increasing the index one by one. We approach the zeroth value from the negative side for the ending point of the flares. This way we obtain the start and end times of all the flares.

Observations

We look at the data collected from Aditya L1 for the month of December. The applied algorithm helps us bin the data, interpolate it, remove the background noise and identify all the flare start and end times. Given below is the data obtained from the application of the algorithm.

Background estimate: 6.587649364670119
Number of flares detected: 25

	Flare peak value	Flare start time	Flare end time	Flare duration (mins)
0	9.641221	6937.412102	6951.512102	14.100000
1	20.062525	25136.492935	25157.626269	21.133333
2	13.376979	25180.009602	25198.792935	18.783333
3	26.162493	25216.626269	25231.959602	15.333333
4	43.295812	25232.026269	25252.142935	20.116667
5	78.158000	25658.842935	25700.292935	41.450000
6	67.329292	25658.842935	25700.292935	41.450000
7	97.244022	25721.276269	25753.792935	32.516667
8	66.609067	25772.742935	25793.576269	20.833333
9	244.764016	25842.326269	25888.092935	45.766667
10	161.688550	25899.159602	25963.992673	64.833071
11	44.619272	25899.159602	25963.992673	64.833071
12	48.414844	25965.559339	25994.876006	29.316667
13	233.584587	26004.976006	26068.642673	63.666667
14	100.199377	26004.976006	26068.642673	63.666667
15	104.891172	26084.726006	26087.526006	2.800000
16	187.715398	26126.592673	26157.009339	30.416667
17	30.202445	26325.276006	26326.659339	1.383333
18	26.840760	26621.609339	26629.109339	7.500000
19	71.843282	26674.392673	26700.992673	26.600000
20	30.674342	26738.109339	26745.159339	7.050000
21	59.717828	26943.176006	26980.576006	37.400000
22	41.217834	27056.309339	27075.492673	19.183333
23	51.894639	27145.559339	27167.809339	22.250000
24	27.892456	27241.042673	27243.342673	2.300000

Figure 2.4.1

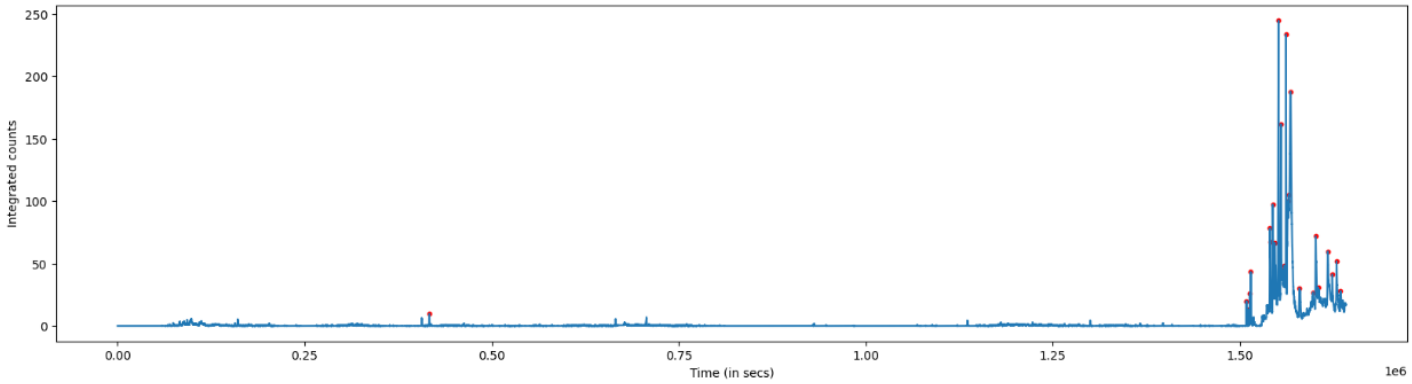


Figure 2.4.2

Fig. 2.4.1 represents the data recorded using the algorithm for the month of December 2019. Fig.2.4.2 represents the same data plotted with time recorded in seconds (relative) and peaks marked with red dots.

3. Curve Fitting

Gaussian rise model

The final step of our automated flare detection system involves fitting a curve to the data points. Curve fitting helps you characterize the features of the data more clearly. In the case of detecting flares, the data may contain sharp peaks or spikes. By fitting a curve (e.g., Gaussian), you can capture key properties of those peaks, such as: Fitting a curve to data points in flare detection helps to discern meaningful patterns, reduce noise, and accurately characterize flare events, allowing for reliable detection and analysis. Curve fitting is essential because it helps you model your data, extract key parameters, and understand the underlying processes in a quantitative and consistent manner. In your flare detection task, it allows you to describe each flare's properties (amplitude, width, decay time), detect trends, and predict future flare behavior based on past data.

We use the model “Gaussian rise and exponential decay” as it can effectively capture the characteristic behavior of a flare's intensity over time. We wish to fit the curve to all the flares detected. The main idea is to fit a curve to each flare's peak and use this fitted curve to model the flare's behavior (e.g., its rise, peak, and decay). We begin by defining the gaussian rise and exponential decay function. It is denoted mathematically as:

$$\mathbf{f\ (t;\ A,}t_o\mathbf{,\sigma,\tau) = }Ae^{\left(\frac{-(t-t_o)^2}{2\sigma^2}\right)}e^{\left(-\frac{(t-t_o)}{\tau}\right)}$$

where the first exponential term stands for the gaussian rise while the second one stands for the decay of the flare. Here, A is the amplitude, σ is the width of FWHM of each flare, and μ is the peak time of each flare, and τ is the decay parameter of the exponential function. We iterate through each flare identified from the data. Firstly, we find the value of tau or the decay time constant. Then we go on to calculate the FWHM of the flares

The Full Width at Half Maximum (FWHM) of a solar flare refers to the duration or the temporal width of the flare's intensity profile, measured at half of its maximum intensity. We set up bounds for the function to achieve more realistic and fit values. It ensures that the optimization process stays within reasonable and physically meaningful limits for the parameters of the model.

We set up a maximum evaluations parameter to ensure that the fitting process doesn't continue indefinitely. If the algorithm exceeds the specified number of function evaluations, it will stop and return the best result it has found so far. This is crucial because it prevents the optimization from running forever, especially when dealing with large datasets or complex models. By setting a reasonable limit, we strike a balance between efficiency and accuracy.

After obtaining the optimized values for the amplitude, peak time, width, and decay time for each flare, we go ahead and fit the curve for the data recorded over the course of one day. This process allows us to quantify each flare's characteristics, such as its intensity, timing, and decay, while ensuring the fitting process remains computationally manageable.

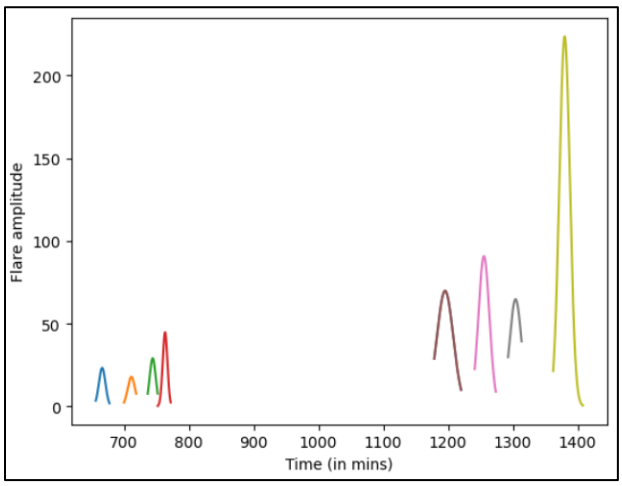


Figure 3.1.1 The plotted curve fit with time from the data collected for December 30th, 2019.

We plot the data we got for each peak with the original data to see how well the curve fits the data.

Flare amplitude	Flare decay constant	Flare peak time	Flare width
23.243065	2634.756986	666.333058	5.048731
Flare amplitude	Flare decay constant	Flare peak time	Flare width
17.831273	266.155901	711.5268	5.689497
Flare amplitude	Flare decay constant	Flare peak time	Flare width
29.074215	135.40977	744.462365	4.708388
Flare amplitude	Flare decay constant	Flare peak time	Flare width
44.748516	91880.696034	763.423805	3.611514
Flare amplitude	Flare decay constant	Flare peak time	Flare width
69.556799	141.133106	1196.637629	12.561005
Flare amplitude	Flare decay constant	Flare peak time	Flare width
69.208534	93.724403	1197.203071	12.560991
Flare amplitude	Flare decay constant	Flare peak time	Flare width
90.737785	176.837052	1255.854157	8.503138
Flare amplitude	Flare decay constant	Flare peak time	Flare width
64.773831	1123.959451	1304.367532	9.276775
Flare amplitude	Flare decay constant	Flare peak time	Flare width
223.533346	690.366367	1380.119159	8.176438

Figure 3.1.2 The optimized values obtained for the peaks detected from the data for 30th December, 2019.

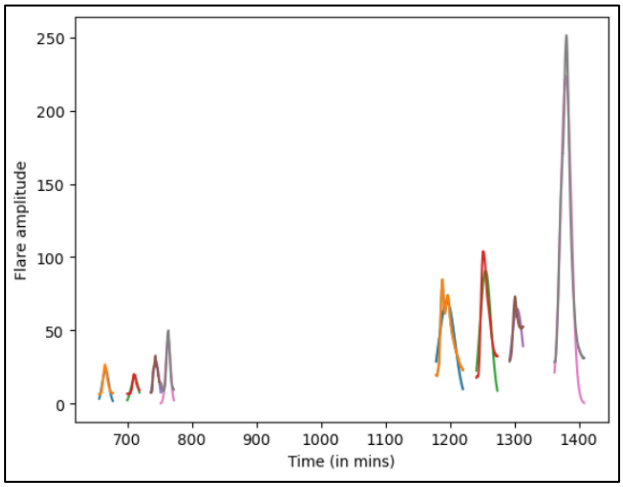


Figure 3.1.3 Optimized curve fits plotted with real world data.

We conclude from this that our curve fit was successful and the Gaussian function accurately represents the data and describes the flare or peak characteristics well. By carefully controlling the optimization process with parameters like maximum evaluations, we ensure that the fitting process remains efficient, reliable, and applicable even to large and complex datasets, such as those recorded over extended periods of time.

4. Conclusion

In this project, we aimed to detect solar flares from time-series data by creating an optimized automated solar flare detection algorithm. We identified different processes and functions we had to use, notified the potential mistakes, and created a setup to help us with the task. We calculated the background estimation and defined a method to find out the start and end times of flares. We then proceeded to fit our peaks to curves using a gaussian rise and exponential decay model. We successfully fitted the curves and got the optimized values for different parameters useful for further analysis. Future work could benefit from incorporating more sophisticated flare models, as well as enhancing the data quality and resolution. This work provides valuable insights into solar flare behavior, and it can be extended for more detailed studies of solar activity and flare detection systems.

References

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